

CS241 REAL ESTATE PROJECT DBMS

Assumptions

- Each property is listed by exactly one agent.
- A property can either be sold once or rented multiple times.
- Clients can act as buyer, seller, tenant, or owner.
- Each locality belongs to a single city (default assumed Mumbai).
- Days on market is calculated externally and stored in the database.
- Property status is maintained as Available, Sold, or Rented.

AI Prompts (Claude)

Prompt 1: "Convert an ER diagram of a real estate system into a normalized relational schema. Include tables for agents, properties, clients, and transactions with proper primary and foreign keys."

Prompt 2: "Enhance the SQL schema by adding constraints such as CHECK, UNIQUE, ENUM, and ensure referential integrity between property, agent, and transaction tables."

Contribution

Gaurav Murali (2401082): Converted ER diagram into relational schema, identified tables and attributes, and defined primary and foreign key relationships.

Mayank Golchha (2401122): Designed and refined AI prompts to generate SQL schema, reviewed AI-generated output, and ensured alignment with project requirements.

Krishna Singh Parmar (2401110): Enhanced schema with constraints such as CHECK, UNIQUE, NOT NULL, ENUM and ensured normalization up to 3NF. Refined transaction tables.

Harsh Upadhyay (2401092): Implemented schema in MySQL, populated tables with sample data, tested queries, and verified referential integrity.

AI Assistance

AI tools were used for generating initial schema drafts, suggesting SQL constraints, and improving structure. Final design, validation, and implementation were completed by the team.